

AWS DMS

Complete Parameters & Settings Cheat Sheet

A comprehensive quick-reference guide for AWS Database Migration Service
Source & Target endpoints · Task settings · Table mappings · Pre-migration
checklist

Oracle

SQL Server

PostgreSQL

MySQL

MongoDB

S3

Redshift

Kinesis

Kafka

DynamoDB

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Tags: **REQ** Required | **REC** Recommended | **OPT** Optional | **CDC** CDC only | **PERF** Performance

§1 Source Endpoint Settings

Oracle Source — Extra Connection Attributes

Parameter	Recommended Value	Tag	Purpose & Notes
useLogminerReader	Y / N	REC	Use LogMiner (Y) or Binary Reader (N). Binary Reader is faster for high-throughput CDC.
useBfile	Y	CDC	Enables BFILE for Binary Reader mode. Required when useLogminerReader=N.
useDirectPathFullLoad	Y	PERF	Uses Oracle Direct Path API for full-load; bypasses redo log, ~3–5× faster.
parallelASMReadThreads	2 – 8	PERF	Threads for reading ASM redo logs in Binary Reader mode (default 2).
readAheadBlocks	150000	PERF	Number of blocks cached ahead. Increase for high-volume tables.
accessAlternateDirectly	false	OPT	Access standby/alternate redo directly; set true for Active Data Guard sources.
useAlternateFolderForOnline	true	OPT	Look in archived log path for online logs if primary path fails.
oraclePathPrefix	/rdsdbdata/db/ORCL	OPT	ASM path prefix for RDS Oracle instances.
usePathPrefix	/rdsdbdata/log	OPT	Log path prefix, required for RDS Oracle with Binary Reader.
replacePathPrefix	true	OPT	Replaces the default path with usePathPrefix value.
retryInterval	5	CDC	Seconds between re-connect retries when redo log unavailable.
archivedLogDestId	1	CDC	LOG_ARCHIVE_DEST_n ID used for reading archived redo logs (1–31).
additionalArchivedLogDestId	2	CDC	Fallback LOG_ARCHIVE_DEST used when primary dest is unavailable.
enableHomogenousLoad	true	PERF	Oracle-to-Oracle homogeneous migration; preserves data types natively.
standbyDelayTime	0	CDC	Minutes behind primary for standby source; compensates for apply lag.
failTasksOnLobTruncation	false	OPT	Fail vs. truncate LOB data exceeding max LOB size. Set true for strict migrations.
numberDataTypeScale	-1	OPT	-1 = float64, 0 = int64; controls how Oracle NUMBER maps to target.
charLengthSemantics	DEFAULT	OPT	CHAR/BYTE/DEFAULT — character length semantics for VARCHAR2 columns.

Oracle CDC Prerequisites: ARCHIVELOG mode ON · Supplemental logging enabled (ALL COLUMNS or at minimum PRIMARY KEY) · LogMiner or Binary Reader access · DMS user needs SELECT_CATALOG_ROLE, SELECT on V\$ARCHIVED_LOG, V\$LOG, V\$LOGFILE, V\$DATABASE, V\$THREAD, DBA_OBJECTS, DBA_SEGMENTS, DBA_TABLES.

SQL Server Source — Extra Connection Attributes

Parameter	Recommended Value	Tag	Purpose & Notes
<code>useBcpFullLoad</code>	<code>Y</code>	PERF	BCP bulk load for full load (faster than ODBC row-by-row).
<code>bcpPacketSize</code>	<code>16384</code>	PERF	BCP network packet size in bytes (4096–65535); larger = faster on LAN.
<code>useThirdPartyBackupDevice</code>	<code>false</code>	OPT	Enable if backup device is 3rd-party (Veritas, LiteSpeed, etc.).
<code>cdcStartPosition</code>	<code>0x000...</code>	CDC	LSN hex string to start CDC from a specific log position.
<code>readBackupOnly</code>	<code>Y</code>	OPT	Read from log backup instead of live transaction log (reduces prod load).
<code>safeguardPolicy</code>	<code>RELY_ON_SQL_SERVER_REPLICATION_AGENT</code>	CDC	Policy for ensuring transactional consistency during CDC. Use <code>EXCLUSIVE_AUTOMATIC_TRUNCATION</code> only in controlled environments.
<code>querySingleAlwaysOnNode</code>	<code>true</code>	OPT	Always query a single AlwaysOn node for CDC; avoids cross-node LSN issues.
<code>controlTablesFileGroup</code>	<code>PRIMARY</code>	OPT	Filegroup for DMS internal control tables on source.
<code>tlogAccessMode</code>	<code>BackupOnly</code>	CDC	ViaFile / BackupOnly — how DMS reads t-log for CDC (RDS SQL Server requires BackupOnly).

SQL Server CDC Prerequisites: MS-CDC (not replication) must be enabled on source DB and each table. Requires sysadmin or db_owner role for setup. RDS SQL Server must use `tlogAccessMode=BackupOnly` with automated backups enabled.

§1 Source Endpoint Settings (continued)

PostgreSQL Source — Extra Connection Attributes

Parameter	Recommended Value	Tag	Purpose & Notes
pluginName	pglogical / test_decoding	REC	pglogical provides better performance; test_decoding is default/fallback.
slotName	dms_replication_slot	CDC	Logical replication slot name. Must pre-exist or DMS creates on start.
heartbeatEnable	true	CDC	Sends heartbeat to advance LSN so slot does not block WAL cleanup.
heartbeatFrequency	5	CDC	Heartbeat interval in minutes (1–60). Lower = more frequent slot advance.
heartbeatSchema	public	CDC	Schema where heartbeat table awsdms_heartbeat is created.
captureDdls	Y	CDC	Capture DDL changes (CREATE TABLE, ALTER TABLE) during CDC. Requires pglogical.
ddlArtifactsSchema	public	CDC	Schema for DDL event trigger functions; must exist and user must have rights.
executeTimeout	100	OPT	Statement timeout in seconds for DMS queries on source.
maxFileSize	1048576	OPT	Max CDC staging file size in KB (default 1 GB). Tune for network throughput.
databaseMode	default	OPT	default or babelfish — use babelfish for Babelfish for Aurora PostgreSQL sources.

MySQL Source — Extra Connection Attributes

Parameter	Recommended Value	Tag	Purpose & Notes
initstmt	SET FOREIGN_KEY_CHECKS=0	REC	Initialization statement run at connect. Disables FK checks for fast full load.
eventsPollInterval	5	CDC	Seconds between binlog polling cycles (default 5); reduce for lower latency.
targetDbType	READ_WRITE	OPT	READ_WRITE or READ_ONLY; applies to target Aurora MySQL.
parallelLoadThreads	1	PERF	Number of parallel threads per table during full load (max 16 for MySQL).
serverTimezone	UTC	OPT	Overrides server timezone; needed when MySQL session TZ differs from UTC.
afterConnectScript	SET sql_mode='STRICT_ALL_TABLES'	OPT	SQL executed after connection established; use for session-level settings.
cdcStartPosition	mysql-bin-change-log.000001:1	CDC	Start CDC from specific binlog position: <filename>:<position>.
BinlogRowImage	FULL	REQ	MySQL server variable — must be FULL for DMS CDC (not MINIMAL/NOBLOB).

MongoDB Source — Extra Connection Attributes

Parameter	Value	Notes
<code>authSource</code>	<code>admin</code>	DB used for authentication.
<code>authMechanism</code>	<code>DEFAULT</code> / <code>SCRAM-SHA-256</code>	Auth protocol; <code>DEFAULT</code> auto-negotiates.
<code>nestingLevel</code>	<code>ONE</code>	<code>NONE</code> =flat JSON string; <code>ONE</code> =expand 1 level of nesting.
<code>extractDocId</code>	<code>true</code>	Extract <code>_id</code> as separate column.
<code>docsToInvestigate</code>	<code>1000</code>	Sample docs for schema inference.
<code>kvPairExtract</code>	<code>false</code>	<code>true</code> = flatten all key-value pairs as columns.
<code>replicaSet</code>	<code>rs0</code>	Replica set name for change stream CDC.

S3 Source — Extra Connection Attributes

Parameter	Value	Notes
<code>bucketFolder</code>	<code>prefix/path</code>	S3 prefix (folder) within bucket.
<code>cdcPath</code>	<code>cdc/</code>	S3 prefix for CDC change files.
<code>csvDelimiter</code>	<code>,</code>	CSV column delimiter character.
<code>csvRowDelimiter</code>	<code>\n</code>	CSV row delimiter.
<code>externalTableDefinition</code>	<code><JSON></code>	Schema definition for source S3 files.
<code>ignoreHeaderRows</code>	<code>1</code>	Skip header rows in CSV files.
<code>rfc4180</code>	<code>true</code>	Enforce RFC 4180 CSV quoting rules.
<code>addColumnNames</code>	<code>true</code>	Add column names to CDC output.

§2 Target Endpoint Settings

MySQL / Aurora MySQL Target — Extra Connection Attributes

Parameter	Recommended Value	Tag	Purpose & Notes
<code>parallelLoadThreads</code>	4 – 8	PERF	Parallel write threads per table during full load. Set <= instance vCPUs.
<code>maxFileSize</code>	512	PERF	Max staging CSV file size in KB before LOAD DATA; 512–8192 is typical.
<code>TargetDbType</code>	READ_WRITE	OPT	Ensures target DB is writable (default); set READ_ONLY only for DR scenarios.
<code>initstmt</code>	SET FOREIGN_KEY_CHECKS=0; SET UNIQUE_CHECKS=0	REC	Disables constraints during load; re-enable manually after.
<code>afterConnectScript</code>	SET sql_mode=''	OPT	Relaxes strict SQL mode to prevent validation errors on migrated data.
<code>cleanSrcMetadataOnMismatch</code>	true	OPT	Clears cached table metadata if schema mismatch detected; prevents stale-schema errors.

PostgreSQL / Aurora PostgreSQL Target — Extra Connection Attributes

Parameter	Recommended Value	Tag	Purpose & Notes
<code>maxFileSize</code>	512	PERF	Max COPY staging file size in KB (default 32 KB); increase for bulk performance.
<code>executeTimeout</code>	100	OPT	Statement timeout (seconds) for DMS-issued statements on target.
<code>heartbeatEnable</code>	true	OPT	Send heartbeat transactions to keep idle connections alive.
<code>databaseMode</code>	default	OPT	default or babelfish; required for Babelfish targets.
<code>BabelfishDatabaseName</code>	mydb	OPT	Target DB name when databaseMode=babelfish.
<code>mapBooleanAsBoolean</code>	true	OPT	Map source BOOLEAN type natively; default is varchar(5) string mapping.
<code>addFullTableName</code>	false	OPT	Prefix columns with schema.table name for disambiguation in UNION scenarios.

Amazon Redshift Target — Extra Connection Attributes

Parameter	Recommended Value	Tag	Purpose & Notes
<code>acceptAnyDate</code>	<code>true</code>	REC	Accepts invalid date formats (e.g., 0000-00-00) without error.
<code>caseSensitiveNames</code>	<code>false</code>	OPT	Redshift is case-insensitive by default; set true only if mixed-case objects used.
<code>compUpdate</code>	<code>false</code>	PERF	Disable automatic compression analysis during COPY (faster full load).
<code>connectionTimeout</code>	<code>5</code>	OPT	Seconds to wait for connection before timeout.
<code>dateFormat</code>	<code>auto</code>	REC	auto detects date format in source data; avoids manual format specification.
<code>emptyAsNull</code>	<code>true</code>	REC	Convert empty strings to NULL; Redshift treats empty strings as empty, not NULL.
<code>encryptionMode</code>	<code>SSE_S3 / SSE_KMS</code>	OPT	Encryption for S3 staging files; SSE_KMS requires <code>serverSideEncryptionKmsKeyId</code> .
<code>fileTransferUploadStreams</code>	<code>10</code>	PERF	Parallel S3 upload streams (1–64). Increase for large tables.
<code>loadTimeout</code>	<code>300</code>	OPT	Max seconds for a single Redshift COPY command before timeout.
<code>maxFileSize</code>	<code>102400</code>	PERF	Max S3 staging file size in KB (default 32 MB). 100 MB often optimal.
<code>replaceInvalidChars</code>	<code><space></code>	OPT	Replace invalid UTF-8 characters instead of failing the COPY.
<code>serverSideEncryptionKmsKeyId</code>	<code>arn:aws:kms:...</code>	OPT	KMS key ARN for SSE_KMS staging encryption.
<code>timeFormat</code>	<code>auto</code>	REC	auto auto-detects time format during COPY.
<code>trimBlanks</code>	<code>true</code>	OPT	Trim trailing spaces from VARCHAR values loaded into Redshift.
<code>truncateColumns</code>	<code>false</code>	OPT	Silently truncate data exceeding column width; use with caution, may lose data.
<code>writeBufferSize</code>	<code>10485760</code>	PERF	Memory buffer for staging files before S3 upload (default 10 MB).

§2 Target Endpoint Settings (continued)

Amazon S3 Target — Key Settings

Parameter	Recommended Value	Tag	Purpose & Notes
<code>bucketFolder</code>	<code>dms-output/</code>	OPT	S3 prefix (virtual folder) within the bucket for output files.
<code>bucketName</code>	<code>my-dms-bucket</code>	REQ	Target S3 bucket name (bucket must already exist).
<code>cannedAclForObjects</code>	<code>private</code>	OPT	Canned ACL applied to written objects. Options: none, private, public-read, etc.
<code>cdcInsertsOnly</code>	<code>false</code>	CDC	Write only INSERTs to S3 (ignore UPDATES/DELETES). Good for append-only lakes.
<code>cdcInsertsAndUpdates</code>	<code>false</code>	CDC	Write only INSERT and UPDATE ops; exclude DELETES.
<code>csvDelimiter</code>	<code>,</code>	OPT	Column delimiter for CSV output (default comma).
<code>csvRowDelimiter</code>	<code>\n</code>	OPT	Row delimiter for CSV output.
<code>csvNullValue</code>	<code>\N</code>	OPT	String representation for NULL values in CSV. Hive/Athena use \N by default.
<code>dataFormat</code>	<code>parquet / csv</code>	REC	parquet = best for analytics (columnar, compressed); csv = max compatibility.
<code>parquetVersion</code>	<code>parquet-2-0</code>	OPT	Parquet format version: parquet-1-0 or parquet-2-0 (2.0 supports more types).
<code>parquetCompressionType</code>	<code>SNAPPY</code>	PERF	SNAPPY (default), GZIP, or NONE. Snappy is fast; GZIP has better ratio.
<code>compressionType</code>	<code>GZIP</code>	PERF	Compression for CSV output: NONE, GZIP, or ZSTD.
<code>datePartitionEnabled</code>	<code>true</code>	REC	Partition S3 output by date (year/month/day) for Athena/ Glue compatibility.
<code>datePartitionSequence</code>	<code>YYYYMMDD</code>	OPT	Date partition folder pattern: YYYYMMDD, YYYYMMDDHH, YYYYMM, MMYYYYDD, DDMMYYYY.
<code>datePartitionDelimiter</code>	<code>SLASH</code>	OPT	Partition folder separator: SLASH (/) or UNDERSCORE (_) or DASH (-).
<code>enableStatistics</code>	<code>true</code>	OPT	Write Parquet statistics; enables predicate pushdown in Athena queries.
<code>encodingType</code>	<code>plain-dictionary</code>	OPT	Parquet encoding: plain, plain-dictionary, or rle-dictionary.
<code>includeOpForFullLoad</code>	<code>true</code>	OPT	Add Op column (I/U/D) to full-load output; enables uniform processing downstream.
<code>maxFileSize</code>	<code>131072</code>	PERF	Max output file size in KB before DMS rotates to a new file (default 32 MB).
<code>rowGroupSize</code>	<code>10240</code>	OPT	Parquet row group size in KB. Larger = better compression; 10–128 MB typical.
<code>serverSideEncryptionKmsKeyId</code>	<code>arn:aws:kms:...</code>	OPT	KMS key for SSE-KMS encryption of S3 objects.
<code>timestampColumnName</code>	<code>dms_timestamp</code>	CDC	Adds a commit-timestamp column to every output row for CDC ordering.
<code>useCsvNoSupValue</code>	<code>false</code>	OPT	Use the csvNoSupValue string for supplemental log columns not present in source.

Amazon Kinesis Target — Key Settings

Parameter	Value	Notes
<code>streamArn</code>	<code>arn:aws:kinesis:...</code>	Target Kinesis stream ARN.
<code>messageFormat</code>	<code>json</code> / <code>json-unformatted</code>	json includes metadata; json-unformatted is compact.
<code>includeTransactionDetails</code>	<code>true</code>	Include transaction begin/commit records.
<code>includePartitionValue</code>	<code>true</code>	Include partition key in message envelope.
<code>partitionIncludeSchemaTable</code>	<code>true</code>	Use schema.table as Kinesis partition key.
<code>includeTableAlterOperations</code>	<code>true</code>	Include DDL ALTER events in stream.
<code>includeControlDetails</code>	<code>true</code>	Include control message metadata.
<code>includeNullAndEmpty</code>	<code>false</code>	Include null/empty column values in JSON output.
<code>noHexPrefix</code>	<code>false</code>	Omit 0x prefix from hex values.

Apache Kafka Target — Key Settings

Parameter	Value	Notes
<code>broker</code>	<code>broker:9092</code>	Comma-separated broker list (host:port).
<code>topic</code>	<code>dms-topic</code>	Default topic name; tables may override.
<code>messageFormat</code>	<code>json</code>	json or json-unformatted output.
<code>messageMaxBytes</code>	<code>1000000</code>	Max Kafka message size in bytes (must match broker max.message.bytes).
<code>partitionIncludeSchemaTable</code>	<code>true</code>	Use schema.table as Kafka message key.
<code>noHexPrefix</code>	<code>false</code>	Omit 0x prefix from binary hex.
<code>securityProtocol</code>	<code>ssl</code> / <code>sasl-ssl</code>	Connection security: plaintext, ssl, sasl-ssl, sasl-plaintext.
<code>saslMechanism</code>	<code>SCRAM-SHA-256</code>	SASL mechanism for auth when securityProtocol=sasl-ssl.
<code>sslCaCertificate</code>	<code><PEM cert></code>	CA cert in PEM for TLS verification.

Amazon DynamoDB Target — Key Settings

Parameter	Recommended Value	Tag	Purpose & Notes
<code>objectMappingType</code>	<code>hashKey</code> / <code>rangeKey</code>	REQ	Defines how source primary key maps to DynamoDB partition (hash) + sort (range) keys.
<code>attributeMappingType</code>	<code>include</code> / <code>exclude</code>	REQ	include lists attributes to map; exclude lists attributes to skip.
<code>writeCapacityUnits</code>	<code>Auto</code> / <code>500</code>	PERF	Provisioned WCUs during full load. Consider PAY_PER_REQUEST and revert after migration.

§3 Task Settings JSON Structure

Task settings are provided as a JSON document when creating or modifying a DMS task (Console → Task → Advanced Settings or via CLI -- replication-task-settings). Below are the most important keys grouped by functional area.

FullLoadSettings

JSON Key	Default / Recommended	Tag	Purpose & Notes
TargetTablePrepMode	DROP_AND_CREATE	REC	DROP_AND_CREATE (default), DO_NOTHING, or TRUNCATE. DROP_AND_CREATE recreates structure; TRUNCATE preserves indexes.
CreatePkAfterFullLoad	false	PERF	Defer PK/index creation until after full load completes. Dramatically speeds up bulk inserts.
StopTaskCachedChangesApplied	false	OPT	Stop task after applying cached CDC changes from full-load phase. Used for cutover orchestration.
StopTaskCachedChangesNotApplied	false	OPT	Stop task before applying cached changes; useful to inspect delta before applying.
MaxFullLoadSubTasks	8	PERF	Max parallel tables loaded simultaneously (1-49). Match to replication instance CPU/network.
TransactionConsistencyTimeout	600	OPT	Seconds to wait for in-flight transactions before starting full load. 600 s default; increase for long transactions.
CommitRate	50000	PERF	Rows per commit batch during full load (default 10,000). Larger batches = faster load, higher memory.

TargetMetadata & LOB Settings

JSON Key	Default / Recommended	Tag	Purpose & Notes
TargetSchema	<schema name>	OPT	Override target schema for all migrated tables.
SupportLobs	true	REC	Enable LOB (BLOB/CLOB/NCLOB) migration. Must be true for any LOB columns.
FullLobMode	false	OPT	false = Limited LOB mode (faster, bounded); true = Full LOB mode (exact, slower, no parallel load).
LobChunkSize	64	OPT	LOB chunk fetch size in KB when FullLobMode=true. Increase for large LOBs (max 102400).
LimitedSizeLobMode	true	REC	Use limited LOB mode; set LobMaxSize to handle the majority of LOB sizes efficiently.
LobMaxSize	32	REC	Max LOB size in KB for limited mode. Data exceeding this is truncated (or fails if FailTaskOnLobTruncation=true).
InlineLobMaxSize	0	PERF	LOBs <= this size (KB) fetched inline (no two-pass). 0=disabled; 32-64 KB can improve CDC throughput.
BatchApplyEnabled	false	PERF	Apply CDC changes in batches rather than row-by-row. Incompatible with LOB columns.

ChangeProcessingTuning — CDC Performance

JSON Key	Default / Recommended	Tag	Purpose & Notes
<code>BatchApplyPreserveTransaction</code>	<code>true</code>	CDC	Preserve transaction atomicity in batch apply mode. Set false only if row order doesn't matter.
<code>BatchApplyTimeoutMin</code>	<code>1</code>	CDC	Min seconds before flushing a batch (1–10800). Lower = more frequent smaller batches.
<code>BatchApplyTimeoutMax</code>	<code>30</code>	CDC	Max seconds to accumulate a batch before forced flush (1–10800).
<code>BatchApplyMemoryLimit</code>	<code>500</code>	CDC	Memory limit in MB for batch change accumulation before a forced flush to target.
<code>BatchSplitSize</code>	<code>0</code>	CDC	Max rows per batch split; 0 = no limit. Set to avoid oversized batches on Redshift/DynamoDB.
<code>MinTransactionSize</code>	<code>1000</code>	CDC	Minimum rows before committing a CDC transaction to target (default 1000). Larger = better throughput.
<code>CommitTimeout</code>	<code>1</code>	CDC	Seconds to wait after MinTransactionSize before forcing commit (1–86400).
<code>MemoryLimitTotal</code>	<code>1024</code>	CDC	Total memory limit in MB for CDC processing. Increase on large instance types.
<code>MemoryKeepTime</code>	<code>60</code>	CDC	Seconds to retain uncommitted changes in memory before spilling to disk.
<code>StatementCacheSize</code>	<code>50</code>	CDC	Prepared statement cache size for CDC apply. Larger reduces parse overhead for many tables.

§3 Task Settings JSON Structure (continued)

StreamBufferSettings

JSON Key	Default / Recommended	Tag	Purpose & Notes
StreamBufferCount	3	PERF	Number of stream buffers (1–8). Increase to 4–8 for high-throughput CDC tasks. Each buffer uses StreamBufferSizeInMB.
StreamBufferSizeInMB	8	PERF	Size of each stream buffer in MB (1–256). Total memory = Count × SizeInMB. Scale up for large transactions.
CtrlStreamBufferSizeInMB	5	OPT	Control stream buffer size in MB (1–32). Used for DDL and control messages.

ErrorBehavior — Error Handling Policies

JSON Key	Default / Recommended	Tag	Purpose & Notes
DataErrorPolicy	LOG_ERROR	REC	Policy when a data conversion error occurs. IGNORE_RECORD / LOG_ERROR / SUSPEND_TABLE / STOP_TASK.
DataTruncationErrorPolicy	LOG_ERROR	OPT	Policy when source data exceeds target column width.
DataErrorEscalationPolicy	SUSPEND_TABLE	OPT	Escalation when DataErrorCount threshold reached: SUSPEND_TABLE or STOP_TASK.
DataErrorEscalationCount	50	OPT	Number of data errors per table before escalation policy triggers.
TableErrorPolicy	SUSPEND_TABLE	REC	Action when a table-level error occurs: SUSPEND_TABLE or STOP_TASK.
TableErrorEscalationPolicy	STOP_TASK	OPT	Escalation after TableErrorEscalationCount is exceeded.
TableErrorEscalationCount	50	OPT	Number of table errors before escalation.
RecoverableErrorCount	-1	REC	-1 = unlimited retries on recoverable (transient) errors. Set positive int for finite retries.
RecoverableErrorInterval	5	OPT	Seconds between retries for recoverable errors.
RecoverableErrorThrottling	true	OPT	Exponentially back off retry interval up to RecoverableErrorThrottlingMax seconds.
RecoverableErrorThrottlingMax	1800	OPT	Max seconds for throttled retry backoff (default 1800 = 30 min).
ApplyErrorDeletePolicy	IGNORE_RECORD	CDC	Policy when DELETE target row not found: IGNORE_RECORD, LOG_ERROR, or STOP_TASK.
ApplyErrorInsertPolicy	LOG_ERROR	CDC	Policy when INSERT conflicts (PK violation): IGNORE_RECORD, LOG_ERROR, UPDATE_RECORD, or STOP_TASK.
ApplyErrorUpdatePolicy	LOG_ERROR	CDC	Policy when UPDATE target row not found.
ApplyErrorEscalationPolicy	LOG_ERROR	CDC	Escalation policy after apply error count threshold exceeded.
ApplyErrorEscalationCount	0	CDC	0 = escalate immediately on any apply error.
FailOnNoTablesCaptured	true	OPT	Fail the task if table mapping matches 0 tables (catches typos in schema/table names).
FailOnTransactionConsistencyBreached	false	OPT	Fail if TransactionConsistencyTimeout exceeded without all open transactions resolved.

Logging — CloudWatch Log Components

Component	Recommended Level	Tag	Purpose & Notes
SOURCE_UNLOAD	LOG_ERROR → DEBUG	OPT	Logs for reading from source during full load.
SOURCE_CAPTURE	LOG_ERROR → DEBUG	CDC	Logs for reading CDC changes from source.
TARGET_LOAD	LOG_ERROR → DEBUG	OPT	Logs for applying data to target during full load.
TARGET_APPLY	LOG_ERROR → DEBUG	CDC	Logs for applying CDC changes to target.
TASK_MANAGER	LOG_ERROR	OPT	Task lifecycle events (start, stop, pause, resume).
TRANSFORMATION	LOG_ERROR → DEBUG	OPT	Table mapping and transformation rule evaluation.
SORTER	LOG_ERROR	CDC	CDC event sorting / ordering buffer operations.
COMMON	LOG_ERROR	OPT	General task-level events.
IO	LOG_ERROR	OPT	I/O layer events (file reads/writes).
FILE_FACTORY	LOG_ERROR	OPT	S3 staging file management events.

Log Levels: DEFAULT (inherits) | LOG_ERROR | LOG_WARNINGS | LOGGER_SEVERITY_DEFAULT | DEBUG | DETAILED_DEBUG. Use DEBUG/DETAILED_DEBUG only during troubleshooting — generates high CloudWatch volume.

Task Settings — JSON Skeleton

```
{
  "TargetMetadata": {
    "TargetSchema": "", "SupportLobs": true,
    "FullLobMode": false, "LobChunkSize": 64,
    "LimitedSizeLobMode": true, "LobMaxSize": 32,
    "InlineLobMaxSize": 0, "BatchApplyEnabled": false
  },
  "FullLoadSettings": {
    "TargetTablePrepMode": "DROP_AND_CREATE",
    "CreatePkAfterFullLoad": false, "MaxFullLoadSubTasks": 8,
    "TransactionConsistencyTimeout": 600, "CommitRate": 50000
  },
  "ChangeProcessingTuning": {
    "BatchApplyPreserveTransaction": true,
    "BatchApplyTimeoutMin": 1, "BatchApplyTimeoutMax": 30,
    "BatchApplyMemoryLimit": 500, "MinTransactionSize": 1000,
    "CommitTimeout": 1, "MemoryLimitTotal": 1024,
    "MemoryKeepTime": 60, "StatementCacheSize": 50
  },
  "StreamBufferSettings": {
    "StreamBufferCount": 3, "StreamBufferSizeInMB": 8,
    "CtrlStreamBufferSizeInMB": 5
  },
  "ErrorBehavior": {
    "DataErrorPolicy": "LOG_ERROR", "RecoverableErrorCount": -1,
    "TableErrorPolicy": "SUSPEND_TABLE",
    "ApplyErrorDeletePolicy": "IGNORE_RECORD",
    "ApplyErrorInsertPolicy": "LOG_ERROR",
    "FailOnNoTablesCaptured": true
  },
  "Logging": {
    "EnableLogging": true, "LogComponents": [
      { "Id": "SOURCE_CAPTURE", "Severity": "LOG_ERROR" },
      { "Id": "TARGET_APPLY", "Severity": "LOG_ERROR" }
    ]
  }
}
```

§4 Table Mappings — Rules Reference

Table mapping rules are a JSON document with a "rules" array. Each rule has rule-type, rule-id, rule-name, object-locator, and type-specific parameters. Rules are evaluated in ascending rule-id order for precedence.

Selection Rules — rule-type: "selection"

Attribute	Values / Options	Tag	Purpose & Notes
rule-action	include / exclude	REQ	include = migrate this object; exclude = skip it. Combine to blacklist specific tables within an included schema.
schema-name	HR / % / HR_*	REQ	Schema pattern. % = wildcard any string; _ = wildcard single char.
table-name	EMPLOYEES / %_HIST / %	REQ	Table pattern. Use % to match all tables in a schema.
filters	[{ "filter-type": "source" ... }]	OPT	Row-level filter. Specify column, filter-operator (ste, gte, eq, between, notNull), and value. Applied at source to limit rows migrated.
load-order	1 - N	OPT	Integer to control table load sequence during full load (lower = first). Useful for dependency ordering.

Transformation Rules — rule-type: "transformation"

rule-action	object-locator target	Tag	Purpose & Example
convert-uppercase	schema / table / column	OPT	Convert schema/table/column name to UPPER CASE. Use for Oracle-to-PostgreSQL migrations.
convert-lowercase	schema / table / column	OPT	Convert name to lower_case. Common for migrating to PostgreSQL (case-insensitive by default).
add-prefix	schema / table / column	OPT	Prepend a string to the object name. E.g., prefix tables with stg_.
remove-prefix	schema / table / column	OPT	Remove a prefix string from object name.
add-suffix	schema / table / column	OPT	Append a string to the object name.
remove-suffix	schema / table / column	OPT	Remove a suffix string from object name.
rename	schema / table / column	OPT	Exact rename using value attribute. E.g., rename schema PROD → DW.
remove-column	column	OPT	Exclude specific columns from migration (e.g., remove audit/system columns).
add-column	table	OPT	Add a computed column to target using expression (e.g., add dms_timestamp column).
change-data-type	column	OPT	Override the default data type mapping for a column. Specify data-type object with type/precision/scale.
define-primary-key	table	OPT	Define PK on target table (useful when source PK is not replicated, e.g., NoSQL sources).
include-column / exclude-column	column	OPT	Granular include/exclude by column name pattern. Alternative to remove-column for selective masking.

Table Mapping — JSON Example (Selection + Transformation)

```
{
  "rules": [
    // 1. Include all tables in HR schema
    { "rule-type": "selection", "rule-id": "1", "rule-name": "include-hr",
      "object-locator": {"schema-name": "HR", "table-name": "%"}, "rule-action": "include" },
    // 2. Exclude staging tables
    { "rule-type": "selection", "rule-id": "2", "rule-name": "excl-staging",
      "object-locator": {"schema-name": "HR", "table-name": "%_STG"}, "rule-action": "exclude" },
    // 3. Rename schema HR → hr (lowercase for PostgreSQL target)
    { "rule-type": "transformation", "rule-id": "3", "rule-name": "lc-schema",
      "rule-action": "convert-lowercase",
      "object-locator": {"schema-name": "%"}, "rule-target": "schema" },
    // 4. Add row filter – migrate only active employees
    { "rule-type": "selection", "rule-id": "4", "rule-name": "active-only",
      "object-locator": {"schema-name": "HR", "table-name": "EMPLOYEES"},
      "rule-action": "include",
      "filters": [{ "filter-type": "source", "column-name": "STATUS",
        "filter-conditions": [{ "filter-operator": "eq", "value": "ACTIVE"}]}]
    }
  ]
}
```

§4 Table Mappings — Parallel Load & LOB Per Table

Parallel Load Rules — rule-type: "table-settings" (parallel-load)

type (value)	Key Parameters	Tag	Purpose & When to Use
<code>none</code>	—	OPT	No parallelism; single-threaded load. Good for small or dependency-critical tables.
<code>partitions-auto</code>	—	PERF	DMS automatically detects partitions (Oracle/MySQL native partitions). Simplest parallel option.
<code>subpartitions-auto</code>	—	PERF	Use Oracle composite subpartitions as parallelism boundaries.
<code>partitions-list</code>	<code>partitions: ["P1", "P2", ...]</code>	PERF	Explicitly list partition names to parallelize. For selective partition migration.
<code>ranges</code>	<code>columns, boundaries</code>	PERF	Split by column value ranges. Ideal for large tables without native partitions. Define boundaries array.

```
// Parallel load using ranges — split large ORDERS table into 4 segments
{
  "rule-type": "table-settings", "rule-id": "10", "rule-name": "parallel-orders",
  "object-locator": {"schema-name": "SALES", "table-name": "ORDERS"},
  "parallel-load": {
    "type": "ranges",
    "columns": ["ORDER_ID"],
    "boundaries": [
      ["1000000"],
      ["2000000"],
      ["3000000"]
    ]
  }
}
```

Boundaries define N+1 segments (3 boundaries → 4 parallel workers: <1M, 1M–2M, 2M–3M, >3M). `MaxFullLoadSubTasks` must be ≥ number of segments.

LOB Per-Table Settings — rule-type: "table-settings" (lob-settings)

Attribute	Recommended Value	Tag	Purpose & Notes
<code>mode</code>	<code>limited</code> / <code>unlimited</code> / <code>none</code>	REQ	<code>limited</code> = <code>LobMaxSize</code> cap; <code>unlimited</code> = exact full LOB; <code>none</code> = exclude LOB columns for this table.
<code>bulk-max-size</code>	<code>16 - 102400</code>	OPT	Max LOB size in KB when <code>mode=limited</code> . Overrides global <code>LobMaxSize</code> for this table only.

```
// Per-table LOB override — large LOBs in DOCUMENTS table, skip LOBs in LOG_EVENTS
[
  {
    "rule-type": "table-settings", "rule-id": "20", "rule-name": "lob-docs",
    "object-locator": {"schema-name": "DMS", "table-name": "DOCUMENTS"},
    "lob-settings": { "mode": "unlimited", "bulk-max-size": "0" }
  },
  {
    "rule-type": "table-settings", "rule-id": "21", "rule-name": "skip-lob-logs",
    "object-locator": {"schema-name": "DMS", "table-name": "LOG_EVENTS"},
    "lob-settings": { "mode": "none" }
  }
]
```


add-column — Built-in Expression Functions

Expression	Returns
<code>\$AR_M_SOURCE_SCHEMA</code>	Source schema name string
<code>\$AR_M_SOURCE_TABLE_NAME</code>	Source table name string
<code>\$AR_M_OPERATION</code>	I / U / D operation type
<code>\$AR_M_SESSION_ID</code>	DMS session identifier
<code>\$AR_M_TIMESTAMP</code>	Commit timestamp (UTC)

```
// add-column example: DMS metadata column
{
  "rule-type": "transformation",
  "rule-action": "add-column",
  "rule-target": "column",
  "object-locator": {
    "schema-name": "%",
    "table-name": "%"
  },
  "value": "dms_op",
  "expression": "$AR_M_OPERATION",
  "data-type": {
    "type": "string",
    "length": 1
  }
}
```

§5 Pre-Migration Checklist

Infrastructure & Networking

- ☐ **VPC & Subnets:** Replication instance in same VPC as target (or VPC-peered to source)
- ☐ **Subnet Group:** DMS Subnet Group created with ≥2 AZ subnets
- ☐ **Security Groups:** Inbound 1521/3306/5432/1433 from replication instance SG to source SG
- ☐ **Replication Instance Class:** r5.xlarge or larger for prod; multi-AZ for HA
- ☐ **Allocated Storage:** ≥50 GB for LOBs or large tables; CDC spill uses local disk
- ☐ **Multi-AZ RI:** Enable for production CDC tasks to survive AZ failure
- ☐ **Publicly Accessible:** Disable unless source/target is outside VPC
- ☐ **IAM Role:** dms-vpc-role and dms-cloudwatch-logs-role exist in IAM
- ☐ **S3 IAM:** dms-s3-role with s3:PutObject/ s3:GetObject on staging bucket
- ☐ **KMS:** Replication instance encrypted; verify CMK key policy if using CMK
- ☐ **Route Tables:** Routes exist from RI subnets to source/target endpoints
- ☐ **DNS Resolution:** Custom DNS resolves source hostname from RI VPC

Source Database Preparation

- ☐ **[Oracle]** ARCHIVELOG mode ON; verify with `SELECT log_mode FROM v$database`
- ☐ **[Oracle]** Supplemental logging: `ALTER DATABASE ADD SUPPLEMENTAL LOG DATA ALL COLUMNS`
- ☐ **[Oracle]** DMS user: `GRANT SELECT_CATALOG_ROLE, CREATE SESSION, SELECT ANY TABLE`
- ☐ **[SQL Server]** MS-CDC enabled: `EXEC sys.sp_cdc_enable_db` on source DB
- ☐ **[SQL Server]** Per-table CDC: `EXEC sys.sp_cdc_enable_table` for each migrated table
- ☐ **[SQL Server]** SQL Server Agent running (required for CDC cleanup job)
- ☐ **[MySQL]** `binlog_format=ROW;`
`binlog_row_image=FULL` in `my.cnf`
- ☐ **[MySQL]** `log_bin=ON;`
`binlog_expire_logs_seconds≥86400`
- ☐ **[PostgreSQL]** `wal_level=logical;`
`max_replication_slots≥2`
- ☐ **[PostgreSQL]** `max_wal_senders≥2;`
`rds.logical_replication=1` for RDS
- ☐ **[PostgreSQL]** Replication slot created (or DMS creates on first start)
- ☐ **[MongoDB]** Replica set mode; oplog retention ≥ migration window
- ☐ **All:** DMS source user has `SELECT` on all migrated tables
- ☐ **All:** Network test: `nc -zv <source-host> <port>` from RI's AZ

Target & Task Preparation

- ☐ **Target Schema:** Target schemas/databases created if `TargetTablePrepMode=DO_NOTHING`
- ☐ **Target User:** DMS target user has `CREATE, INSERT, UPDATE, DELETE, DROP`
- ☐ **[Redshift]** Staging S3 bucket exists; dms-redshift-role can `COPY` from bucket
- ☐ **[DynamoDB]** Tables created (or DMS creates if IAM allows `dynamodb:CreateTable`)
- ☐ **[Kinesis/Kafka]** Stream/topic created; DMS IAM role has `kinesis:PutRecord`
- ☐ **Schema Conversion:** Run AWS SCT; resolve all action items before task start
- ☐ **Table Mapping:** JSON validated; test against source using DMS test endpoint
- ☐ **Endpoint Test:** "Test Connection" passes for both source & target endpoints
- ☐ **LOB Strategy:** Identify all LOB columns; choose limited vs. unlimited per table
- ☐ **Parallel Load:** Identified large tables (>1 GB) for ranges/partition parallel load
- ☐ **Row Counts:** Baseline row counts captured for post-migration validation
- ☐ **Validation:** Enable Data Validation in task settings for automated row count diff
- ☐ **Monitoring:** CloudWatch alarms on `CDCLatencySource`, `CDCLatencyTarget`
- ☐ **Rollback Plan:** Define RPO/RTO; have reverse-sync task ready for cutover failure
- ☐ **Maintenance Window:** Disable RI auto-upgrade during active CDC tasks

Quick Decision: LOB Mode Selection

Scenario	Mode	LobMaxSize	BatchApply?	Trade-off
No LOB columns	SupportLobs=false	N/A	Yes	Fastest option; enable batch apply for max CDC throughput.
Small LOBs (<32 KB typical)	LimitedSizeLobMode	32	No	Good balance; truncates oversized LOBs silently unless FailOnLobTruncation=true.
Medium LOBs (32 KB–1 MB)	LimitedSizeLobMode	1024	No	Larger LobMaxSize increases memory pressure; monitor RI memory.
Large/variable LOBs (>1 MB)	FullLobMode	N/A (LobChunkSize=64+)	No	Exact copy; significant performance impact; no parallel full load.
Mixed (some tables with LOBs)	Per-table lob-settings	Table-specific	No for LOB tables	Use table-settings to apply unlimited mode only to LOB tables.

DMS Task Type Selection Guide

Task Type	Use Case	Parallel Load?	Key Consideration
Full load only	One-time migration, no ongoing sync	Yes	Fastest; no CDC overhead; application downtime required at cutover.
CDC only	Ongoing replication from an existing copy	No	Requires starting SCN/LSN from an already-migrated baseline snapshot.
Full load + CDC	Live migration with minimal downtime	Yes	DMS caches CDC changes during full load then applies; recommended for most prod migrations.

Performance Tips: 1) Use r5.xlarge+ for tables >50 GB or >1000 TPS CDC. 2) Set MaxFullLoadSubTasks=8 and increase for wide tables. 3) Enable batch apply (BatchApplyEnabled=true) for high-volume CDC to non-LOB targets. 4) For Redshift targets, increase fileTransferUploadStreams to 20 and maxFileSize to 100 MB. 5) Use parquet format for S3 targets to reduce storage and improve Athena performance. 6) Monitor CDCLatencySource and CDCLatencyTarget CloudWatch metrics; target < 30 sec.

Common Pitfalls: 1) Forgetting to enable supplemental logging on Oracle source (task starts but CDC produces corrupt rows). 2) Using FullLobMode with parallel load — DMS silently drops parallel load, severely impacting full-load time. 3) Replication slot bloat on PostgreSQL source — always enable heartbeatEnable=true. 4) SQL Server tlogAccessMode — RDS SQL Server MUST use BackupOnly; on-prem can use ViaFile. 5) DynamoDB provisioned WCUs — always scale up before migration, down after.